

Roll No. ....

Total Pages : 3

**CMCA/M-22**  
**MACHINE LEARNING**

**24531**

Paper–MCA-20-44

Option–(ii)

Time Allowed : 3 Hours]

[Maximum Marks : 75

**Note** : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

**Compulsory Question**

1. (a) List the applications of machine learning.
- (b) Distinguish between supervised learning and Reinforcement learning. Illustrate with an example.
- (c) Write down the major differences between K-means clustering and hierarchical clustering.
- (d) What is significance of optimal separating hyperplane in SVM?

**UNIT–I**

2. (a) What are important objectives of machine learning? What are the basic design issues and approaches to machine learning? Discuss.
- (b) What do you understand by decision tree? Discuss the decision tree algorithm of learning.

3. (a) Discuss the characteristic of the problem for decision tree learning.
- (b) What do you mean by a well-posed learning problem? Explain the important features that are required to well-define a learning problem.

**UNIT–II**

4. (a) What are Bayesian Belief nets? Where are they used?
- (b) What is linear regression? Discuss the procedure of finding the line of best fit using linear regression.
5. (a) Explain Naive Bayes Classifier with an example.
- (b) What is Bayes theorem? What are the pre-requisites for it? Discuss its use in machine learning.

**UNIT–III**

6. (a) Discuss in detail the different steps in expectation maximization algorithm.
- (b) What is hierarchical clustering? What is the purpose of it? Discuss.
7. What do you understand by clustering? Write a detailed note on k-means clustering. What are its limitations? Discuss.

**UNIT–IV**

8. (a) What do you understand by Dimensionality Reduction? Explain the principal component analysis for dimensionality reduction.

- (b) What is recommender system? Discuss the use of collaborative filtering in recommender system. Use suitable examples.
- 9. (a) What do you understand by anomaly detection? What is reinforcement learning? Discuss the use of reinforcement learning in anomaly detection.
- (b) What is SVM? What are the pros and cons associated with it? Discuss.